

Exercise 4 - Analysis Results

F1 — OneMax

- **Ordering:** $(1+1) \text{ EA} \leq \text{RLS} \leq \text{MMAS}(\rho = 1) \leq \text{MMAS}^*(\rho = 1) \leq \text{GA} \gg \text{RS}$.
- **Dynamics:** Rapid early gains for $(1+1) \text{ EA/RLS}$; MMAS with $\rho = 1$ quickly saturates pheromones and catches up; $\rho = 1/\sqrt{n}$ and $\rho = 1/n$ adapt more slowly.
- **Variance:** Narrow across all non-random methods once near optimum; RS markedly flatter and noisier.

F2 — LeadingOnes

- **Ordering:** $\text{RLS} > (1+1) \text{ EA} \geq \text{MMAS}^*(\rho = 1/\sqrt{n}) \geq \text{MMAS}(\rho = 1/\sqrt{n}) \geq \text{GA} \gg \text{RS}$.
- **Dynamics:** Position-dependent progress rewards single-bit moves; $\rho = 1$ tends to over-commit early and can stall, while $\rho = 1/\sqrt{n}$ balances learning and avoids premature saturation; $\rho = 1/n$ is robust but slower within the budget.
- **Variance:** Wider than F1; MMAS^* (strict acceptance) is typically a bit steadier than MMAS .

F3 — BinaryValue

- **Ordering:** $(1+1) \text{ EA} \approx \text{RLS} \approx \text{MMAS/MMAS}^*(\rho = 1) \geq \text{GA} \gg \text{RS}$.
- **Dynamics:** Strict monotonicity favors aggressive exploitation; $\rho = 1$ performs best among ACO settings; smaller ρ slows convergence without improving robustness in practice.
- **Variance:** Generally small; curves cluster tightly for non-random methods.

F18 — LABS (rugged/plateaus)

- **Ordering:** $\text{MMAS/MMAS}^*(\rho = 1/\sqrt{n}) \geq \text{GA} \geq (1+1) \text{ EA} \geq \text{RLS} \gg \text{RS}$.
- **Dynamics:** Reinforcement towards good templates (ACO) and recombination (GA) help traverse plateaus; $\rho = 1$ often saturates too fast and can get stuck; $\rho = 1/n$ adapts too slowly to capitalize within the fixed budget.
- **Variance:** Broad $\pm 1\sigma$ bands reflecting landscape ruggedness; MMAS^* slightly reduces lateral drift.

F23

- **Ordering:** $\text{GA} \approx \text{MMAS/MMAS}^* (\rho = 1/\sqrt{n}) \geq (1+1) \text{EA} \geq \text{RLS} \gg \text{RS}$.
- **Dynamics:** Balanced ρ provides the best trade-off between exploration and exploitation; $\rho = 1$ occasionally shows early gains but more frequent stagnation; $\rho = 1/n$ typically under-explores.
- **Variance:** Moderate to high; GA shows wider spread but higher ceilings.

F24

- **Ordering:** $\text{MMAS/MMAS}^* (\rho = 1/\sqrt{n}) \geq \text{GA} \geq (1+1) \text{EA} \geq \text{RLS} \gg \text{RS}$.
- **Dynamics:** ACO benefits from template consolidation while retaining stochastic bit sampling; strict acceptance (MMAS*) slightly improves stability on long plateaus.
- **Variance:** Noticeably wider than unimodal cases; medians track means well for ACO with $\rho = 1/\sqrt{n}$.

F25

- **Ordering:** $\text{GA} \approx \text{MMAS/MMAS}^* (\rho = 1/\sqrt{n}) \geq (1+1) \text{EA} \geq \text{RLS} \gg \text{RS}$.
- **Dynamics:** Problem structure favors recombination; ACO with $\rho = 1/\sqrt{n}$ keeps exploring while leveraging accumulated pheromones; $\rho = 1$ prone to premature convergence; $\rho = 1/n$ too conservative for the budget.
- **Variance:** Highest among the set; best-runs gap is largest for GA and ACO.

Cross-function takeaways

- **Evaporation (ρ) is the performance dial:** $\rho = 1$ excels on easy unimodal tasks (F1/F3) but risks early lock-in on rugged/plateaued problems; $\rho = 1/\sqrt{n}$ is a strong default across mixed landscapes; $\rho = 1/n$ is robust but usually too slow within the fixed budget.
- **Acceptance rule matters:** MMAS ($\geq$) moves laterally more and can progress faster early; MMAS* ($>$) trades a bit of speed for stability on plateaus.
- **Relative to baselines:** RLS/(1+1) EA dominate F1/F3; ACO ($\rho = 1/\sqrt{n}$) and GA tend to lead on F18/F23/F24/F25; RS is consistently weakest.
- **Variance:** Rugged instances show broader spreads; ACO with $\rho = 1/\sqrt{n}$ and MMAS* reduce churn without sacrificing much peak performance.